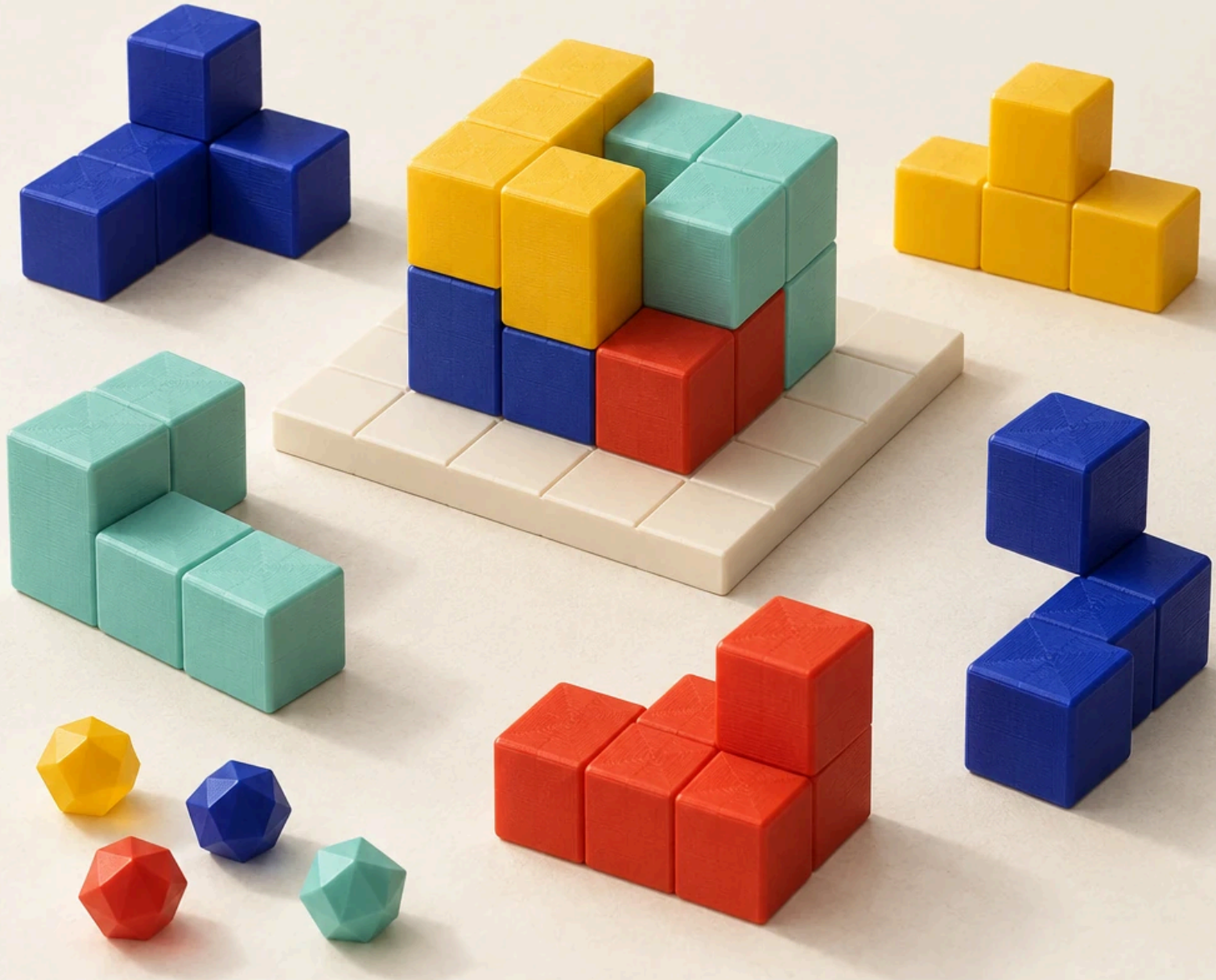
A FAMILY 3D PUZZLE GAME

WONGO

CLASSIC EDITION

Play online
wongo.duras.me



Good things fit together.

Family rulebook · English edition

2–8 players / 9 rounds / 2 layers

WONGO 3D

FAMILY RULEBOOK · ENGLISH · VERSION 2.0 · PHYSICAL D6 CARDS

2–8 players · 9 rounds · Ages 10+ suggested · About 25–40 minutes

Build on your printed card before time runs out. Everyone plays at once and earns gems. The website is optional for solutions and practice; you do not need it at the table. Wongo is an independent family adaptation inspired by Ubongo 3D Master, not an official product.

What you need

- The shared pool of 16 numbered shapes: **40 pieces for 2–4 players or 80 for 5–8**, in the manifest quantities. These are not personal complete sets. Cube pitch is 20 mm, with 0.5 mm edge chamfers.
- **Double-sided A5 D6 cards**, a normal six-sided die, timer, opaque gem bag and score paper. The download has **12 numbered packs of 8 cards**. Easy is on the front, hard on the back. Each side has one footprint and six piece lists.
- **90 gems:** 18 red, 27 blue, 18 green and 27 amber. One set works for every player count.

Gem	Points
Red	4
Blue	3
Green	2
Amber	1

Set up

1. Shuffle **complete numbered packs**, keep their cards together and choose nine packs for nine rounds. Never deal unrelated cards: inventories are checked per pack and die face.
2. For **2–4 players**, use only slots 1–4 (block A) or only 5–8 (block B) from one pack. Give each player a different card from that block. For **5–8**, use slots 1–N from one pack and the 80-piece pool. Smaller groups still fit.
3. Choose **one common difficulty**. Easy uses three different colours; hard four pieces, one of each colour. Turn all cards to that side. The inventory guarantee does not cover mixed difficulties, mixed packs, or mixed blocks with 40 pieces.
4. Use **two minutes per round**, or a longer agreed common limit. Give beginners a practice round.
5. Display **nine blue and nine amber bonus gems**, one pair per round. Bag the rest: initially **18 of each colour**. Keep solutions hidden.

The building challenge

The single footprint on your card applies to **all six die rows**. The row changes the pieces, not the board. Fill **every square exactly two cubes high**.

- Use every listed piece once and add no extra pieces.
- Rotate pieces freely in three dimensions; do not alter or mirror their shapes.
- Leave no gaps or overlaps; nothing may project outside the footprint or above layer two.
- The final build must stand on the table. You may lift and rearrange pieces while solving.

Shallow grooves show the individual cubes. Judge the 20 mm lattice, not tiny decorative chamfer gaps. Colours and IDs identify pieces; board cells do not specify colours.

Play a round

1. Deal cards from the next complete pack as described above. **Roll once for everybody.** Each player finds that die face on their card and takes exactly its listed pieces. Do not roll separately per player.
2. Check the pieces, then start the shared timer. Solve your own footprint. On finishing, say **“Wongo!”**, leave the build in place and record the order. Others continue until time expires.
3. Check finished builds before awarding gems. A mistake earns nothing unless fixed before the time limit. A displayed reference solution does not prove the player's build is correct.
4. Every correct finisher draws **one random gem**. The fastest also takes the round's **blue** bonus; the second-fastest takes **amber**.
5. Return unawarded bonuses to the bag and all pieces to the pool. Put the complete pack aside and use the next one. Do not retain pieces or mix packs.

For an indistinguishable finish tie, draw lots for bonus order; all correct finishers still draw a gem. If **nobody** finishes, you may allow **one second chance**: restart the same timer once on the same cards and die result. Agree beforehand. If nobody then solves, return both bonuses to the bag. The round still counts; do not add rounds.

Win

After round nine, total the gem points. Highest wins; **a tied highest score is a shared victory**. Gem count alone does not decide. Change the common level or timer between games to suit the group.

Use the virtual puzzle library

Search the **W3 code beside the chosen die row**, not the D6 pack ID. It opens that piece list and a reference solution. The printed footprint may be rotated to share a common orientation; turn the card when comparing it with the screen.

Choose **Try this puzzle** for practice with X/Y/Z rotations and two layer grids. The anchor is the lower corner of the piece's bounding box and may be empty. **+X points right; +Y points up.** Illegal overlaps and bounds are blocked. Any complete legal solution is accepted, even if different from the saved example. Partial practice builds may temporarily have unsupported cubes.

Reveal the solution after the competitive round. Revealing it during a round makes that player's puzzle ineligible for rewards. Without WebGL, use numbered 2D layers. The website does not synchronize timers, rolls, finish order or scores.

Every printed alternative is independently solved, and every common die result checked against the block's shared pool. This proves geometry and inventory availability, not printer tolerances or drop resistance.

Printing and care

Download the **D6 physical cards**. Use A4 landscape, **100% / actual size**, no fitting, margins, headers or footers, and duplex **flip on the short edge**. The 96-page file uses **48 A4 sheets** and yields 96 double-sided A5 cards. Back slots are intentionally reversed. Measure the **40 mm guide**, check one sheet, then cut at the centre. Keep packs and blocks together. Do not print solutions on the backs.

Models use 20 mm cubes, 0.5 mm chamfers, 0.1 mm layers, 3 walls and a 5% gyroid starting profile. Print a small fit sample first: low infill and printer settings need a physical test.

Solo practice

Take any card, choose a side, roll and time yourself. No gems or multiplayer inventory checks are needed. Keep small pieces and gems away from children who might put them in their mouths.

Reference: Albi's Ubongo 3D Master rulebook. Wongo's D6 packs, player counts, puzzles, inventory scheduling and 90-gem setup are custom adaptations.